Valence Alignment Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Contrasts on the Exchange-Aligned Data Model

PENPAL Project

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Contents

This notebook compares **valence alignment** across the three experimental conditions using the new exchange-aligned structure.

- **Human-AI**: Human co-writing with an LLM
- **Human-Human**: Two humans co-writing together
- **AI-AI**: Two AI agents co-writing together

Important structural note:

- `author_1` / `author_2` are now treated as **slot 1 / slot 2** in the cleaned exchange sequence.
- In **Human-AI**, those slots also map onto human vs AI roles.
- In **Human-Human** and **AI-AI**, the slots are positional sides of the exchange, not stable identity labels.

The analysis window is restricted to **complete exchanges** and the default aligned turn range (`analysis_turn` 1 to 9).

Primary cross-condition object. Each story contributes one unsigned, label-invariant dyad asymmetry magnitude:

$$A_{\text{valence}}(i) \;=\; \left|\,\text{atanh}(r_{1\to 2,i}) - \text{atanh}(r_{2\to 1,i})\,\right|$$

where $r_{1\to 2,i}$ is the per-story Pearson correlation between same-turn slot 1 and slot 2 valence, and $r_{2\to 1,i}$ is the correlation between slot 2's previous-turn valence and slot 1's current-turn valence. The main question is whether A_{valence} differs across conditions. The directional-slope LMMs are retained as supplemental views. Starter-mechanism and Human-AI speaker-type decompositions are deferred to a follow-up pass.

Article-use note. The primary cross-condition claim should use the unsigned per-story A_{valence} test below. Signed directional slot models are diagnostic only: in Human-Human and AI-AI, Slot 1 / Slot 2 are positional labels rather than theoretically meaningful agent identities.

1 Load Data

```
## Data availability:
```

```
## # A tibble: 6 x 4
##   file                                    `human-ai` `human-human` `ai-ai`
##   <chr>                                   <lgl>      <lgl>         <lgl>
## 1 dyadic_sentiment_scores.parquet         TRUE       TRUE          TRUE
## 2 interaction_level_surface_metrics.parquet TRUE     TRUE          TRUE
## 3 novelty_scores.csv                      TRUE       TRUE          TRUE
## 4 semantic_exploration_binned.parquet     TRUE       TRUE          TRUE
## 5 story_embeddings_interaction_level.parquet TRUE    TRUE          TRUE
## 6 interaction_level_stories_filtered.csv  TRUE       TRUE          TRUE
```

Table 1: Loaded exchange-aligned valence data by condition

condition	n_stories	n_turns	min_turn	max_turn
Human-AI	100	792	1	8
Human-Human	36	324	1	9
AI-AI	80	720	1	9

2 Preprocessing

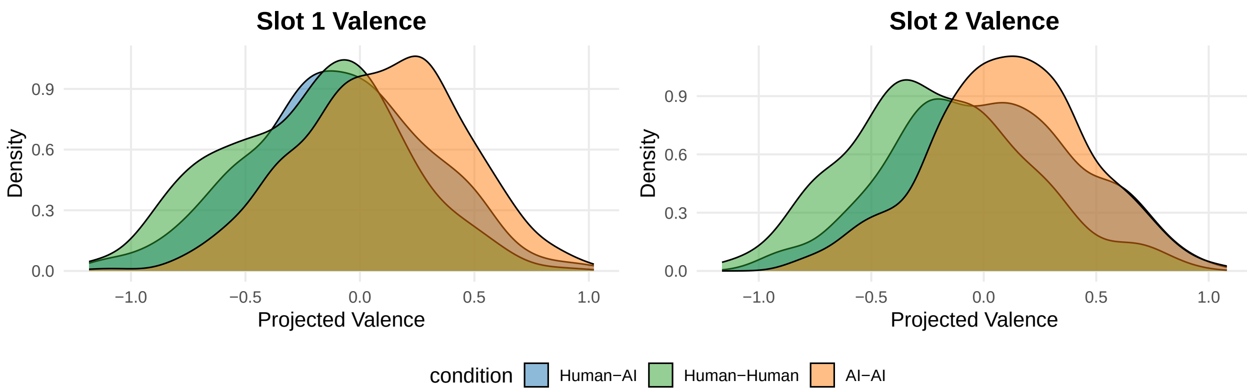
```
## Analysis rows: 3240
```

```
## Stories: 216
```

3 Descriptive Checks

Table 2: Mean projected valence by condition

condition	n_stories	n_turns	slot_1_mean	slot_2_mean
Human-AI	100	692	-0.093	0.016
Human-Human	36	288	-0.208	-0.207
AI-AI	80	640	0.093	0.112



4 Primary: Per-Story Asymmetry Magnitude

```
## Stories with both directional correlations available:
```

Table 3: Per-story A_valence sample sizes by condition

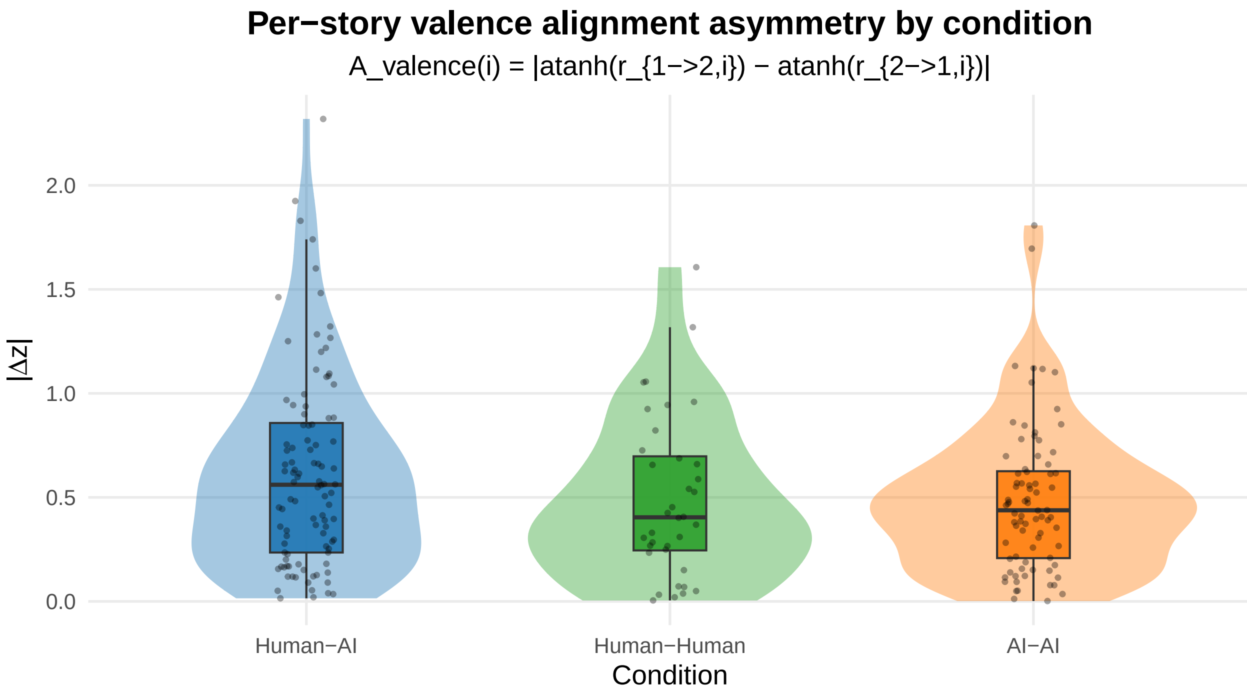
condition	n
Human-AI	100
Human-Human	36
AI-AI	80

Table 4: Per-story |Fisher-z asymmetry| by condition

condition	n	mean_A	sd_A	median_A
Human-AI	100	0.618	0.471	0.561
Human-Human	36	0.495	0.392	0.404
AI-AI	80	0.485	0.353	0.438

```
## One-way ANOVA: A_valence ~ condition

##              Df Sum Sq Mean Sq F value Pr(>F)
## condition     2   0.91  0.4542   2.602 0.0765 .
## Residuals   213  37.18  0.1745
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Kruskal-Wallis (robust): A_valence ~ condition

##
##  Kruskal-Wallis rank sum test
##
## data:  A_valence by condition
## Kruskal-Wallis chi-squared = 3.6397, df = 2, p-value = 0.162

##
## Pairwise Wilcoxon (BH-adjusted):

##
##  Pairwise comparisons using Wilcoxon rank sum test with continuity correction
```

```
##
## data:  df_asym$A_valence and df_asym$condition
##
##             Human-AI Human-Human
## Human-Human 0.29     -
## AI-AI       0.24     0.85
##
## P value adjustment method: BH
```



5 Bootstrap CIs and Effect Sizes on `A_valence`

Table 5: Bootstrap 95% CI on per-condition mean of A_valence (R = 10,000)

condition	n	mean	ci_lower	ci_upper
Human-AI	100	0.618	0.529	0.712
Human-Human	36	0.495	0.369	0.623
AI-AI	80	0.485	0.413	0.564

Table 6: Pairwise Cliff's delta on A_valence

contrast	n_a	n_b	cliff_delta	magnitude
Human-AI vs Human-Human	100	36	0.147	negligible
Human-AI vs AI-AI	100	80	0.152	small
Human-Human vs AI-AI	36	80	-0.022	negligible

6 Planned Contrast: HA vs (HH + AA)/2

Table 7: Planned contrast: A_valence in HA vs the average of HH and AA

contrast	estimate	se	t	df	p	ci_lower	ci_upper
Human-AI vs (Human-Human + AI-AI)/2	0.128	0.061	2.113	160.366	0.036	0.008	0.248

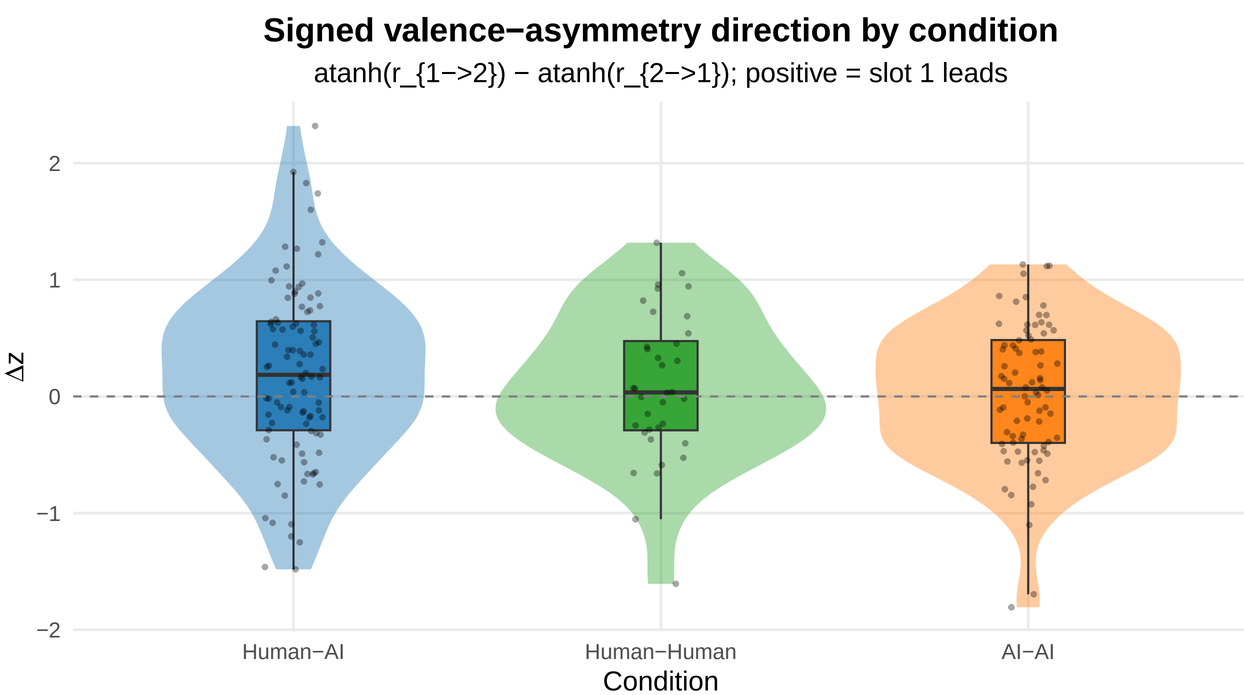
7 Directional Stability of Signed Valence Asymmetry

Tests whether the *direction* of the slot-1 vs slot-2 alignment difference is consistent across stories within each condition. The signed quantity is $\text{signed}_i = \text{atanh}(r_{1\to2,i}) - \text{atanh}(r_{2\to1,i})$. HA-specific identity-tied directionality predicts high `stability` and `sign_consistency` in HA, low in HH and AA, even when unsigned `A_valence` is similar.

Table 8: Per-condition directional stability of signed valence asymmetry

condition	n	mean	sd	stability	sign_consistency	one_sample_t_p
Human-AI	100	0.199	0.753	0.265	0.580	0.009
Human-Human	36	0.082	0.631	0.130	0.528	0.441
AI-AI	80	0.024	0.602	0.039	0.550	0.727

```
## Brown-Forsythe variance-equality p = 0.1314
```



8 Permutation Test on `A_valence ~ condition`

Table 9: Permutation test on A_valence ~ condition (R = 10,000 story-label permutations)

F_obs	p_perm	R
2.6021	0.0755	10000

9 Summary

Primary cross-condition result: per-story $A_{\text{valence}} = |\,\text{atanh}(r_{1\to 2,i}) - \text{atanh}(r_{2\to 1,i})|$, tested with ANOVA + Kruskal and pairwise BH-adjusted Wilcoxon.

Robustness layers added:

- bootstrap 95% CIs on per-condition mean of A_{valence}
- pairwise Cliff's δ effect sizes
- planned single-df contrast HA vs (HH+AA)/2
- 10k story-label permutation test on $A_{\text{valence}} \sim$ condition

The Human-AI signed Human/AI directional decomposition lives in analysis/human-ai/valence_alignment_analysis.Rmd and is not duplicated here.

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized asymmetry across all conditions)

Analysis completed: 2026-05-05 12:45:22.926937